

LAVYA JAIN

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Portfolio

LinkedIn

GitHub

EDUCATION

Dr. D.Y. Patil Institute of Technology, Pune

B.E – Information Technology CGPA: 8.14/10

Aug 2023 – July 2027

Pune, Maharashtra

St. Joseph's Academy, Dehradun

Class XII (ISC) – 90.2%

2022

Dehradun, Uttarakhand

St. Joseph's Academy, Dehradun

Class X (ICSE) – 91.2%

2020

Dehradun, Uttarakhand

PROJECTS

Sanrakshan CTC | C++17, Crow, WebSockets, JavaScript | [GitHub](#) | [Demo](#)

Jan 2026 – Mar 2026

- Engineered a multithreaded railway traffic control simulator where every train executes independently using `std::thread`, coordinated through mutex-protected shared track resources.
- Implemented collision avoidance using `non-blocking try_lock()`, eliminating hold-and-wait deadlocks while dynamically rerouting trains through graph-based path planning.
- Built a live WebSocket telemetry pipeline streaming real-time updates to multiple concurrent train simulations with sub-second latency.
- Implemented an interactive deadlock simulation demonstrating Dining Philosophers-style resource contention and synchronization using mutexes and non-blocking locks.

VAJRA | C++, React, Three.js, Node.js, ZeroMQ, WebSockets | [GitHub](#) | [Demo](#)

Mar 2026 – Apr 2026

- Architected a cyber-physical power grid simulator that models electrical load propagation, cascading failures, and infrastructure attacks through a deterministic C++ simulation engine.
- Developed a ZeroMQ-to-WebSocket bridge enabling bidirectional communication between a native C++ engine and React frontend with sub-second updates.
- Built a 3D digital twin using React Three Fiber with real-time telemetry overlays, threat visualization, AI voice alerts, and operator-focused SCADA interfaces.
- Implemented simulated industrial cyberattacks including False Data Injection, Packet Spoofing, and Volumetric DDoS with automated anomaly detection and node isolation strategies inspired by MITRE ATT&CK for ICS.

VeriFind | React Native, Node.js, MongoDB, Solidity, IPFS | [GitHub](#) | [Demo](#)

Aug 2025 – Feb 2026

- Designed a blockchain-backed ownership verification platform that establishes immutable ownership history for physical electronic devices using ERC-721 smart contracts.
- Implemented a gasless meta-transaction relayer allowing users to interact with blockchain assets without holding cryptocurrency while preserving cryptographic ownership verification.
- Built a React Native mobile application integrated with MongoDB and IPFS for real-time device registration, ownership transfer, and decentralized metadata storage.
- Developed secure ownership validation that restricts sensitive actions such as transfer, lost-device reporting, and recovery exclusively to the wallet cryptographically linked to the registered device.

TECHNICAL SKILLS

Languages: C++, JavaScript, TypeScript, Java, SQL

Core Concepts: Data Structures & Algorithms, Operating Systems, Computer Networks, Multithreading, Concurrent Programming, Distributed Systems, Blockchain

Frameworks & Libraries: React, React Native, Node.js, Express.js, Three.js, React Three Fiber, Solidity, Ethers.js, Crow, Tailwind CSS

Developer Tools: Git, GitHub, Linux, Docker, ZeroMQ, WebSockets, Postman, Vercel

Databases & Storage: MongoDB, PostgreSQL, IPFS

ACHIEVEMENTS

- Winner, **Hash-It-Out Hackathon 2026**, Dr. D.Y. Patil Institute of Technology.
- Finalist, **Hackathon Vortexa 2025**, presenting an AI-powered financial strategy platform.
- Presented Sanrakshan CTC to judges through a live demonstration of concurrent train scheduling, collision avoidance, and deadlock visualization.